CHEMISTRY

1. Which of the following ion shows spin only magnetic moment of 4.9 B.M.?

(1) Mn^{+2}

(2) Cr^{+2}

(3) Fe^{+3}

(4) Co^{+2}

Ans. (2)

Sol. [Unpaired electron]

$Mn^{+2} \rightarrow 3d^5$ (n = 5)

$Cr^{+2} \rightarrow 3d^4$ n = 4

$Fe^{+3} \rightarrow 3d^5$ n = 5

$Co^{+2} \rightarrow 3d^7$ n = 3

$\mu = \sqrt{n(n+2)}$

2. Which of the following has highest atomic number?

(1) Po

(2) Pt

(3) Pr

(4) Pb

Ans. (1)

Sol. Theoretical

3. An ideal gas with an adiabatic exponent 1.5, initially at 27°C is compressed adiabatically from 800cc to 200 cc. The final temperature of the gas is

(1) 600 K

(2) 300 K

(3) 450 K

(4) 273 K

Ans. (1)

Sol. $PV^{\gamma} = \text{cont.}$

$TV^{\gamma-1} = \text{cont.}$

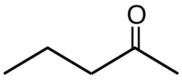
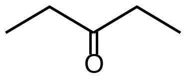
$T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$

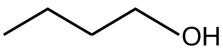
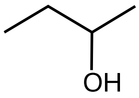
$$T_2 = T_1\left(\frac{V_1}{V_2}\right)^{\gamma-1}$$

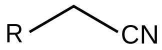
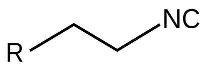
$$T_2 = 300\left(\frac{800}{200}\right)^{\frac{1}{2}}$$

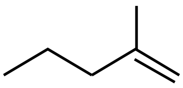
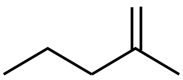
$T_2 = 600K$

4. Which of the following is/are correct

(a)  and  : are Metamers

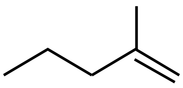
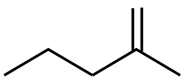
(b)  and  : are Position isomers

(c)  and  : are Functional isomers

(d)  and  : are Position isomers

(1) (a), (b), (c), (d)
(2) (a), (b), (d)
(3) (a), (b)
(4) (a), (b), (c)

Ans. (4)

Sol.  and  : are identical.

5. Which is correct

(1) $A + e^- \rightarrow A^- \rightarrow$ is always exothermic
(2) $A \rightarrow A^+ + e^- \rightarrow$ is always endothermic.
(3) IE_1 of (Be) < IE_1 (B)
(4) Lithium is most electropositive in its group.

Ans. (2)

Sol. Factual.

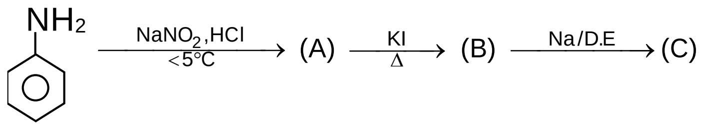
6.

List-I	List-II
(a) $Ni(CO)_4$	(P) sp^3d
(b) $PtCl_4^{2-}$	(Q) sp^3
(c) PF_5	(R) sp^3d^2
(d) SF_6	(S) dsp^2

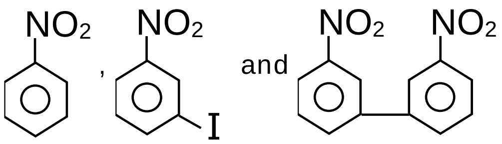
(1) (a) – q, (b) – s, (c) – p, (d) - r
(2) (a) – q, (b) – s, (c) – r, (d) - p
(3) (a) – s, (b) – q, (c) – p, (d) - r
(4) (a) – s, (b) – q, (c) – r, (d) - p

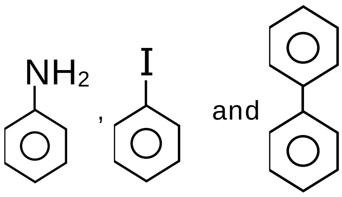
Ans. (1)

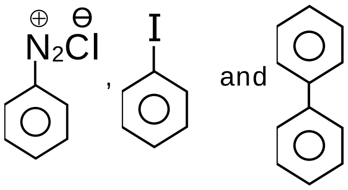
Sol. (a) $Ni(CO)_4 \rightarrow sp^3$
(b) $PtCl_4^{2-} \rightarrow dsp^2$
(c) $PF_5 \rightarrow sp^3d$
(d) $SF_6 \rightarrow sp^3d^2$

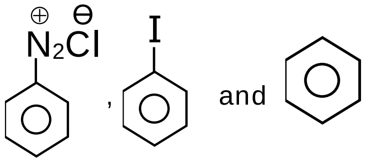
7. 

A, B & C will be ?

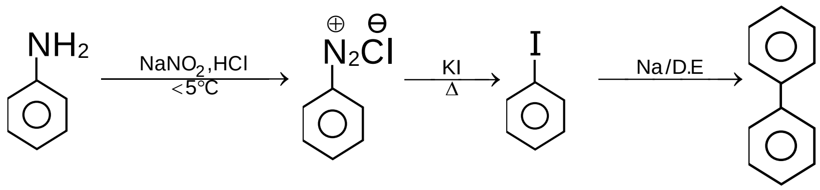
(1) 

(2) 

(3) 

(4) 

Ans. (3)

Sol. 

8. 0.5 gram of an organic compound gives 1.46 gram CO_2 and 0.9 gram H_2O. What is the percentage of Carbon in organic sample. [Nearest Integer]

Ans. 80

Sol. $C_xH_y\left(x+\frac{y}{4}\right)O_2 \rightarrow xCO_2+\frac{y}{2}H_2O$

$\because$ 44g CO_2 contains $\rightarrow$ 12g C

So 1.46g CO_2 contains $\rightarrow \frac{12}{44}\times 1.46 = 0.398$g

$\%\text{ of C} = \frac{0.398}{0.5} \simeq 80$

9. $PCl_5 \rightleftharpoons PCl_3 + Cl_2$

At const. volume and temp, Xenon is added

(1) Amount of PCl_3 will increase

(2) Amount of PCl_3 will Decrease

(3) Amount of PCl_3 and Cl_2 will remain constant.

(4) Amount of PCl_5 will Increase

Ans. (3)

Sol. No effect in the composition after addition of inert gas at constant volume.

10. 2 moles each of ethylene glycol and glucose are mixed with 500g of water. Find the boiling point of solution given $K_b = 0.52K\ Kg\ Mol^{-1}$

(1) 377.16K

(2) 368.84K

(3) 376.16K

(4) 369.84K

Ans. (1)

Sol. $\Delta T_b = K_b \times m$

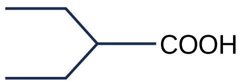
$= 0.52 \times \frac{4}{0.5}$

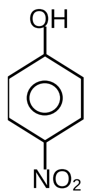
= 4.16

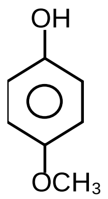
Boiling Temp = 373 + 4.16

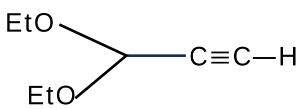
= 377.16 K.

11. Which one of the following is more acidic than others?

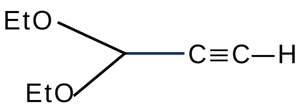
(1) 

(2) 

(3) 

(4) 

Ans. (4)

Sol. 

12. Order of limiting molar conductivites of the following cations at 298K is $H^+, Ca^{+2}, Na^+, K^+, Mg^{2+}$

(1) $Na^+ < K^+ < Mg^{2+} < Ca^{2+} < H^+$

(2) $Mg^{2+} < Ca^{2+} < K^+ < Na^+ < H^+$

(3) $Mg^{2+} < Na^+ < Ca^{2+} < H^+ < K^+$

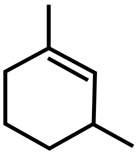
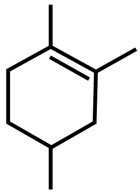
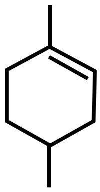
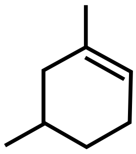
(4) $Mg^{2+} < K^+ < Ca^{2+} < H^+ < Na^+$

Ans. (1)

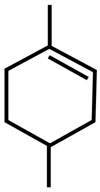
Sol. Charge$\uparrow$ Molar conductivity$\uparrow$

Molar Conductivity $\alpha \dfrac{1}{\text{Hydrated Ionic radii}}$

13. Which of the following will give 3-methyl-6-oxoheptanal as the product of reductive ozonolysis

(1) 
(2) 
(3) 
(4) 

Ans. (3)

Sol.  $\xrightarrow[Zn-H_2O]{O_3} CH_3-\overset{\overset{O}{\parallel}}{C}-CH_2-CH_2-\underset{\underset{CH_3}{|}}{CH}-CH_2-\overset{\overset{O}{\parallel}}{C}-H$

14. 0.42 gram of an organic compound containing 2 nitrogen atom per molecule is reacted to produced N_2 gas. Find volume (ml) of N_2 formed at NTP (molar mass of organic compound = 84)

Ans. 112

Sol. $2\times\dfrac{0.42}{84} = 2\times$ moles of Nitrogen

Vol. of N_2 at STP = $\dfrac{1}{200}\times 22.4 = 0.112l$

= 112 ml

15.

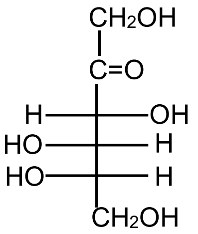
$$MCl + K_2Cr_2O_7 \xrightarrow{H^+} X$$
$$X \xrightarrow{NaOH} Y$$
$$Y \xrightarrow{H^+} Z + H_2O$$

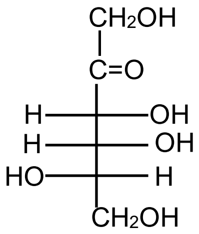
Calculate Number of terminal O-atoms present In compound z.

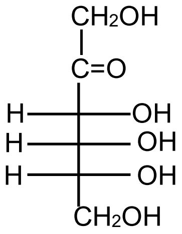
Ans. 6

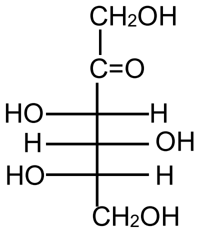
Sol. NCERT

16. Perfect L-fructose structure will be ?

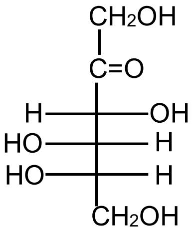
(1) 

(2) 

(3) 

(4) 

Ans. (1)

Sol. 

L-fructose

17. Correct order of wavelength absorb of visible light by the following complexes

$$\underset{(I)}{\left[Ni(CN)_6\right]^{-4}}, \underset{(II)}{\left[Ni(NH_3)_5 H_2O\right]^{2+}}, \underset{(III)}{\left[Ni(NH_3)_5 Cl\right]^{+}} \underset{(IV)}{\left[Ni(NH_3)_6\right]^{+2}}$$

(1) (I) < (II) < (III) < (IV)

(2) (I) < (IV) < (II) < (III)

(3) (III) < (II) < (IV) < (I)

(4) (II) < (III) < (I) < (IV)

Ans. (2)

Sol. $\Delta_0 = \dfrac{hc}{\lambda_{absorbed}}$

18. Statement-I: N—N has less bond strength than P—P.

Statement-II: All group-15 elements in +3 oxidation state undergo disproportionation.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and Statement-II are false

(2) Statement-I is false but Statement-II is true

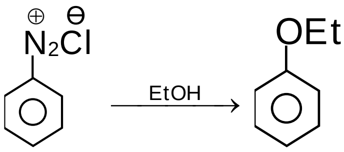
(3) Both Statement I and Statement-II are true

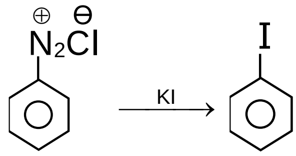
(4) Statement-I is true but Statement-II is false

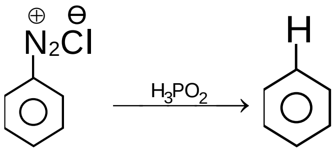
Ans. (4)

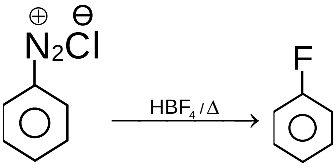
Sol. $>\ddot{N}-\ddot{N}< \; < \; >P-P<$ (Order of bond strength)

19. Which of the following is incorrect reaction product

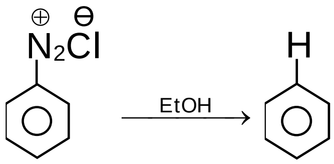
(1) 

(2) 

(3) 

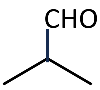
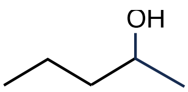
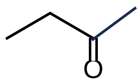
(4) 

Ans. (1)

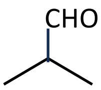
Sol. 

20. Which can’t give iodoform test

ethanol, Isopropyl alcohol, petan-2-one, petan-3-one, butanal, pentanal

Ans. 4

Sol. petan-3-one, butanal, pentanal,  can’t give iodoform test.

21. Which of the following property show irregular trend in group 16?

(1) Electronegativity (2) Atomic Radius

(3) Electron Affinity (4) Ionisation Enthalpy

Ans. (3)

Sol. Electron Affinity order of group 16

S > Se > Te > O

22. Which of the following statement(s) is/are incorrect

I. NO_2 dimerises easily

II. NF_5 does not exists but PF_5 exists.

III. The oxides N_2O_5 and P_2O_5 are purely acidic but As_2O_5 and Sn_2O_5 are basic.

IV. Nitorgen cannot form $d\pi$-$p\pi$ bond as the heavier elements can.

(1) Only I, II and IV (2) Only III (3) Only III and IV (4) Only I and II

Ans. (2)

Sol. Conceptual